1/12/2015

Green's functions for some stationary differential equations

Poisson's eq for scalar potential $\phi(r)$

$$\nabla^2 \phi(r) = -\frac{1}{\varepsilon_0} \rho(r)$$

$$\phi(r) = \int d^3\vec{r}' \frac{\rho(\vec{r}')}{4\pi\varepsilon_0 |\vec{r}-\vec{r}'|}$$

For a point charge q at $\vec{r}'$, $\rho(\vec{r}) = q\delta(\vec{r}-\vec{r}') \Rightarrow \nabla^2\phi = -\frac{q}{\varepsilon_0}\delta(r-r')$

Green's function (GF) $\nabla_r^2 G(r-r') = \delta(\vec{r}-\vec{r}')$

Then $\phi(\vec{r}) = \cancel{\int d^3\vec{r}' G(\vec{r}-\vec{r}') \rho(\vec{r}')}$

$$\nabla_r^2 \phi(\vec{r}) = \cancel{\int d^3\vec{r}' \nabla^2 G(\vec{r}-\vec{r}') \rho(\vec{r}')} = \int d^3\vec{r}' \left(-\frac{4\pi}{\varepsilon_0}\delta(\vec{r}-\vec{r}')\right)\rho(\vec{r}')$$

In $d=3$ $\nabla_r^2 G(\vec{r},\vec{r}') = \delta(\vec{r}-\vec{r}') \Rightarrow G(\vec{r},\vec{r}') = -\frac{1}{4\pi|\vec{r}-\vec{r}'|}$

$$\phi(\vec{r}) = \int d^3\vec{r}' G(\vec{r},\vec{r}')\left[-\frac{\rho(\vec{r}')}{\varepsilon_0}\right] = \int d^3\vec{r}' \frac{\rho(\vec{r}')}{4\pi\varepsilon_0|\vec{r}-\vec{r}'|}$$

Fourier transformation:

$$G(\vec{R}) = \int \frac{d^3\vec{k}}{(2\pi)^3} e^{i\vec{k}\cdot\vec{R}} G(\vec{k})$$

$$G(\vec{k}) = \int d^3\vec{R}\, e^{-i\vec{k}\cdot\vec{R}} G(\vec{R})$$

$$\Rightarrow \nabla_R^2 G(\vec{R}) = \delta(\vec{R}), \quad \vec{R} = \vec{r}-\vec{r}'$$

$$\int \frac{d^3\vec{k}}{(2\pi)^3}(-\vec{k}^2) e^{i\vec{k}\cdot\vec{R}} G(\vec{k}) = \int \frac{d^3\vec{k}}{(2\pi)^3} e^{i\vec{k}\cdot\vec{R}}$$

$$G(\vec{k}) = -\frac{1}{k^2}$$

$$G(\vec{R}) = -\int \frac{d^3\vec{k}}{(2\pi)^2} \frac{e^{i\vec{k}\cdot\vec{R}}}{k^2} = -\int_0^{2\pi} d\varphi \int_0^{\pi} \frac{d\theta \sin\theta}{(2\pi)^3} \int_0^{\infty} dk\, k^2 \frac{e^{i\vec{k}\cdot\vec{R}}}{k^2}$$

$$= -\frac{1}{(2\pi)^2} \int_{-1}^{1} d(\cos\theta) \int_0^{\infty} dk\; e^{ikR\cos\theta} =$$

$$= -\frac{1}{(2\pi)^2} \int dk \left. \frac{e^{ikR\cos\theta}}{ikR} \right|_{-1}^{1} = -\frac{1}{(2\pi)^2} \int_0^{\infty} dk\; \frac{2i\sin(kR)}{ikR} =$$

$$= -\frac{1}{2\pi^2} \int_0^{\infty} \frac{d(kR)}{R} \frac{\sin(kR)}{kR} = -\frac{1}{2\pi^2} \frac{1}{R} \left(\frac{\pi}{2}\right) = -\frac{1}{4\pi R}$$

GF for time-independent Schrodinger eqn

$$i\hbar \frac{\partial \psi(x,t)}{\partial t} = \hat{H}\,\psi(x,t) \quad , \hat{H} = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + U(x) = \hat{H}_0 + U(x)$$

$$\psi(x,t) = \psi(x)\, e^{-i\frac{Et}{\hbar}} \Rightarrow E\,\psi(x) = \hat{H}\,\psi(x)$$

GF: $G(x,x')$ is the solution of

$$(E-\hat{H})\,G(x,x') = \delta(x-x')$$

let us find $G_0(x,x')$: $(E-\hat{H}_0)\,G_0(x,x') = \delta(x-x')$

$$\left(E + \frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}\right) G_0(x,x') = \delta(x-x')$$

For bound states (to be studied later) $E<0$

$$G(x,x') = \begin{cases} A(x')\, e^{\varkappa(x-x')} + B(x')\, e^{-\varkappa(x-x')} & x<x' \\ C(x')\, e^{-\varkappa(x-x')} + D(x')\, e^{\varkappa(x-x')} & x>x' \end{cases}$$

where $\frac{\hbar^2\varkappa^2}{2m} = -E > 0$, $\varkappa = \sqrt{\frac{-2mE}{\hbar^2}} > 0$

Boundary Conditions (BC) : $G(x\to\pm\infty, x') = 0$

$$\Rightarrow B=0,\ D=0$$

Integrate $\int_{x'-0}^{x'+0} dx \left(E + \frac{\hbar^2}{2m}\frac{d^2}{dx^2}\right) G(x,x') = \int_{x'-0}^{x'+0} dx\, \delta(x-x') = 1$

$$\Rightarrow \frac{d}{dx}G(x=x'+0,x') - \frac{d}{dx}G(x=x'-0,x') = \frac{2m}{\hbar^2}$$

$$\Rightarrow -\varkappa C - (\varkappa A) = \frac{2m}{\hbar^2}$$

$$A + C = \frac{-2m}{\hbar^2\varkappa}$$

$$G(x = x'+0, x') = G(x = x'-0, x') \quad \text{(GF itself is continuous)}$$

$$\Rightarrow \quad A = C \quad \Rightarrow \quad A = \frac{-m}{\varkappa \hbar^2}$$

$$G(x, x') = \frac{-m}{\varkappa \hbar^2} e^{-\varkappa |x - x'|}$$

Thus, general solution of the equation $(E<0)$

$$E\psi(x) + \frac{\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2} = f(x)$$

can be written as $\psi(x) = \underbrace{a e^{-\varkappa x} + b e^{\varkappa x}}_{\substack{\text{solution of} \\ \text{homogeneous eqn (with } f=0)}} + \int_{-\infty}^{\infty} G(x,x') f(x') dx'$

Here: $a = b = 0$ due to BC for $\psi(x)$

For the Schrodinger eqn

$$-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \psi(x) + U(x)\psi(x) = E\psi(x)$$

$$\Rightarrow \quad f(x) = U(x)\psi(x)$$

$$\Rightarrow \quad \psi(x) = \int_{-\infty}^{\infty} G(x,x') U(x') \psi(x') dx'$$

$$= \frac{-m}{\varkappa \hbar^2} \int_{-\infty}^{\infty} dx' \, e^{-\varkappa |x - x'|} U(x') \psi(x')$$

Suppose arbitrary potential $U(x) \leq 0$, $U(x) \to 0$ $x \to \pm\infty$

Consider the ground state w.f. $\psi_0(x)$: it has no zeros and can be chosen $\psi_0(x) > 0$.

let $x = x_0$ where x_0 is the point of maximum for $\psi_0(x)$

$$\psi_0(x_0) = \frac{-m}{\varkappa \hbar^2} \int_{-\infty}^{\infty} dx' \, e^{-\varkappa |x_0 - x'|} U(x') \psi_0(x') =$$

$$= \frac{m}{\varkappa \hbar^2} \int_{-\infty}^{\infty} dx' \, e^{-\varkappa |x_0 - x'|} \psi_0(x') \, |U(x')| \leq$$

$$\leq \frac{m}{\varkappa \hbar^2} \int dx' |U(x')| \, \psi_0(x_0)$$

because by construction $\psi_0(x_0) \geq e^{-\varkappa |x_0 - x'|} \psi_0(x')$

$$\Rightarrow \quad 1 \leq \frac{m}{\varkappa \hbar^2} \int dx' \, |U(x')|$$

but $\varkappa = \sqrt{\dfrac{-2mE_0}{\hbar^2}} = \sqrt{\dfrac{2m|E_0|}{\hbar^2}}$

$$\frac{2m|E_0|}{\hbar^2} \leq \frac{m^2}{\hbar^4} \left(\int_{-\infty}^{\infty} dx' \, U(x') \right)^2$$

$$|E_0| \leq \frac{m}{2\hbar^2} \left(\int_{-\infty}^{\infty} U(x) \, dx \right)^2 .$$

$$\psi(x) = -\frac{m}{\varkappa\hbar^2}\int_{-\infty}^{\infty} e^{-\varkappa|x-x'|}\, U(x')\, \psi(x')\,dx'$$

1) $U(x) = -\alpha\,\delta(x)$

$$\psi(x) = \frac{m\alpha}{\varkappa\hbar^2}\, e^{-\varkappa|x|}\,\psi(0) \Rightarrow \psi(0) = \frac{m\alpha}{\varkappa\hbar^2}\,\psi(0)$$

$$\varkappa = \frac{m\alpha}{\hbar^2}$$

$\Rightarrow \psi(x) = \psi(0)\, e^{-\varkappa|x|}$ and $\psi(0)$ is fixed by normalization.

2) $U(x) = -\alpha\left(\delta(x-a) - \delta(x+a)\right) \Rightarrow \int_{-\infty}^{\infty} dx\, U(x) = 0.$

$$\psi(x) = \frac{m\alpha}{\varkappa\hbar^2}\int dx'\left(\delta(x'-a) - \delta(x'+a)\right) e^{-\varkappa|x-x'|}\,\psi(x')$$

$$= \frac{m\alpha}{\varkappa\hbar^2}\left(e^{-\varkappa|x-a|}\,\psi(a) - e^{-\varkappa|x+a|}\,\psi(-a)\right)$$

$$\psi(a) = \frac{m\alpha}{\varkappa\hbar^2}\left(\psi(a) - e^{-2\varkappa a}\,\psi(-a)\right) = \frac{1}{z}\left(\psi_a - e^{-2\varkappa a}\psi_{-a}\right)$$

$$\psi(-a) = \frac{m\alpha}{\varkappa\hbar^2}\left(e^{-2\varkappa a}\,\psi(a) - \psi(-a)\right) = \frac{1}{z}\left(e^{-2\varkappa a}\psi_a - \psi_{-a}\right)$$

$$\cancel{e^{\varkappa a}\psi_a = \frac{m\alpha}{\varkappa\hbar^2}\left(e^{\varkappa a}\psi_a - e^{-\varkappa a}\psi_{-a}\right)}$$

$$\cancel{e^{\varkappa a}\psi_{-a} = \frac{m\alpha}{\varkappa\hbar^2}\left(e^{-\varkappa a}\psi_a - e^{\varkappa a}\psi_{-a}\right)}$$

let $\frac{\varkappa\hbar^2}{m\alpha} = z$, $\varkappa = \frac{m\alpha}{\hbar^2} z$

$$(z-1)\,\psi_a = -e^{-2\varkappa a}\,\psi_{-a} = -e^{-2\varkappa a}\,\frac{e^{-2\varkappa a}}{z+1}\,\psi_a$$

$$(z+1)\,\psi_{-a} = e^{-2\varkappa a}\,\psi_a$$

$$\frac{z^2 - 1 + e^{-4\varkappa a}}{z+1}\,\psi_a = 0$$

$$z^2 = 1 - e^{-4\varkappa a} = 1 - e^{-\frac{4m\alpha a}{\hbar^2} z}$$

let $\beta = \frac{4m\alpha}{\hbar^2} z$, $z = \frac{\hbar^2}{4m\alpha}\beta$

$$\left(\frac{\hbar^2 \beta}{4m\alpha}\right)^2 = 1 - e^{-\beta}$$ $\Rightarrow$ <u>there always is a solution</u>

$\Rightarrow$ <u>bound state</u>

$$z \approx \sqrt{1 - e^{-\frac{4m\alpha a}{\hbar^2}}}$$ for $\frac{m\alpha a}{\hbar^2} \geq 1$ (delta function far apart $\Rightarrow$ just a bound state at attractive one)

$$z \approx 1 - \frac{e}{2}^{-\frac{4m\alpha a}{\hbar^2}}$$

For $\frac{m\alpha a}{\hbar^2} \ll 1$: $z^2 \approx 1 - \left(1 - \frac{4m\alpha a}{\hbar^2} z\right) = \frac{4m\alpha a}{\hbar^2} z$

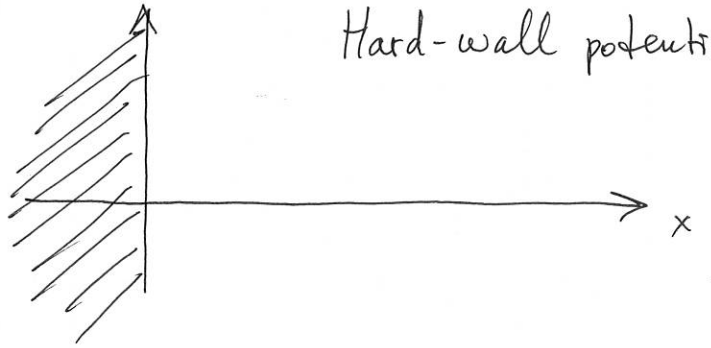
$$z \approx \frac{4m\alpha a}{\hbar^2}$$ (closely spaced δ-functions)

$$z = \frac{\varkappa \hbar^2}{m\alpha} \approx \frac{4m\alpha a}{\hbar^2}$$

$$\varkappa = \frac{4m^2\alpha^2}{\hbar^4} a \Rightarrow \varkappa a = \left(\frac{2m\alpha a}{\hbar^2}\right)^2 \ll 1$$

$$E_0 = -\frac{\hbar^2 \varkappa^2}{2m} = -\frac{\hbar^2}{2ma^2}(\varkappa a)^2 = -\frac{\hbar^2}{2ma^2}\left(\frac{2ma\alpha}{\hbar^2}\right)^4 = -\frac{8\alpha^4 m^3 a^2}{\hbar^6}$$

1/14/2015



Hard-wall potential : $U(x) = \begin{cases} 0, & x > 0 \\ \infty, & x < 0 \end{cases}$

Find GF for $E < 0$.

Clearly, $\psi(x=0) = 0$.

$$(E - \hat{H}_0)\, G_0(x, x') = \delta(x - x')$$

but boundary conditions are (BC) : $\begin{cases} G(x=0, x') = 0 \\ G(x \to \infty, x') = 0 \end{cases}$

<u>Long way</u>

$$G_0(x, x') = \begin{cases} A e^{\varkappa(x-x')} + B e^{-\varkappa(x-x')} & x < x' \\ C e^{-\varkappa(x-x')} + D e^{\varkappa(x-x')} & x > x' \end{cases}$$

$$G_0(x=0, x') = 0 \Rightarrow A e^{-\varkappa x'} + B e^{\varkappa x'} = 0$$

$$B = -A e^{-2\varkappa x'}$$

$$G_0(x \to +\infty, x') = 0 \Rightarrow D = 0.$$

Thus, $$G_0(x, x') = \begin{cases} A\left(e^{\varkappa(x-x')} - e^{-2\varkappa x'} e^{-\varkappa(x-x')}\right), & x < x' \\ C e^{-\varkappa(x-x')}, & x > x'. \end{cases}$$

But we have 2 more conditions:

$$G_0(x'+0, x') = G_0(x'-0, x') \Rightarrow A\left(1 - e^{-2\varkappa x'}\right) = C$$

$$\frac{d}{dx} G_0(x'+0, x') - \frac{d}{dx} G_0(x'-0, x') = \frac{2m}{\hbar^2}$$

$$-\varkappa C - \left(\varkappa A\left(1 + e^{-2\varkappa x'}\right)\right) = \frac{2m}{\hbar^2}$$

$$\Rightarrow \quad C + A\left(1+e^{-2\varkappa x'}\right) = -\frac{2m}{\varkappa\hbar^2}$$

$$A\left(1-e^{-2\varkappa x'}\right) + A\left(1+e^{-2\varkappa x'}\right) = -\frac{2m}{\varkappa\hbar^2}$$

$$A = -\frac{m}{\varkappa\hbar^2}$$

$$\Rightarrow \quad B = +\frac{m}{\varkappa\hbar^2}e^{-2\varkappa x'}, \quad C = -\frac{m}{\varkappa\hbar^2}\left(1-e^{-2\varkappa x'}\right)$$

$$G_0(x,x') = \begin{cases} -\frac{m}{\varkappa\hbar^2}\left(e^{\varkappa(x-x')} - e^{-\varkappa(x+x')}\right), & x<x' \\ -\frac{m}{\varkappa\hbar^2}\left(1-e^{-2\varkappa x'}\right)e^{-\varkappa(x-x')}, & x>x' \end{cases}$$

Since $|x-x'| = \begin{cases} x-x' & \text{for } x>x' \\ x'-x & \text{for } x<x' \end{cases}$, we have

$$G_0(x,x') = -\frac{m}{\varkappa\hbar^2}\begin{cases} e^{-\varkappa|x-x'|} - e^{-\varkappa|x+x'|}, & x<x' \\ e^{-\varkappa|x-x'|} - e^{-\varkappa|x+x'|}, & x>x' \end{cases}$$

So that finally $G_0(x,x') = -\frac{m}{\varkappa\hbar^2}\left(e^{-\varkappa|x-x'|} - e^{-\varkappa|x+x'|}\right)$

for x <u>and</u> x' ~~on~~ positive x-axis

(and zero otherwise).

<u>Short way</u>: place image source at $x=-x'$, i.e. $-\delta(x+x')$.

The difference form $G_0(x,x') = -\frac{m}{\varkappa\hbar^2}\left(e^{-\varkappa|x-x'|} - e^{-\varkappa|x+x'|}\right)$

satisfies all required conditions for <u>$x, x' > 0$</u>.

$\Rightarrow$ hence it represents correct GF.

Now, let us put attractive delta-function potential at $x=a$: $-\alpha\,\delta(x-a)$.

$\Rightarrow$ Schrodinger eqn reads

$$E\psi(x) + \frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} = U(x)\psi(x) = -\alpha\,\delta(x-a)\psi(x)$$

$$\Rightarrow \quad \psi(x) = \int_0^\infty dx'\; G_0(x,x')\,U(x')\,\psi(x') =$$

$$= -\frac{m}{\varkappa\hbar^2}(-\alpha)\int_0^\infty dx'\left(e^{-\varkappa|x-x'|} - e^{-\varkappa(x+x')}\right)\delta(x'-a)\,\psi(x')$$

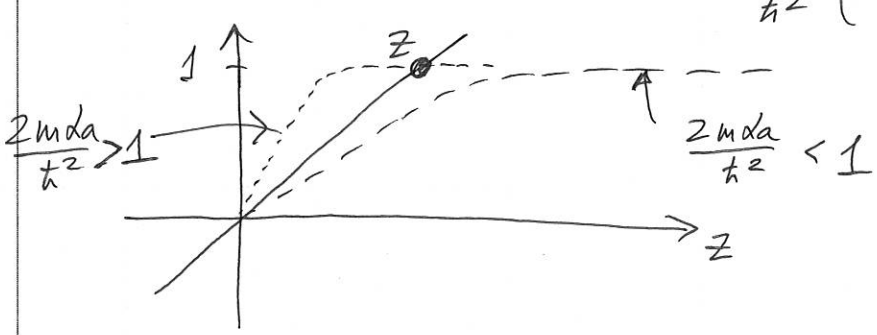
$$= \frac{m\alpha}{\varkappa\hbar^2}\left(e^{-\varkappa|x-a|} - e^{-\varkappa(x+a)}\right)\psi(a).$$

Check: $\psi(x=0)=0$ <u>ok</u>

$$\psi(a) = \frac{m\alpha}{\varkappa\hbar^2}\left(1-e^{-2\varkappa a}\right)\psi(a)$$

$\Rightarrow$ eigenvalue eqn: $1 = \dfrac{m\alpha a}{\hbar^2(\varkappa a)}\left(1-e^{-2\varkappa a}\right)$

let $z=\varkappa a \quad \Rightarrow \quad z = \dfrac{m\alpha a}{\hbar^2}\left(1-e^{-2z}\right) \Rightarrow$ solution appears if $\dfrac{2m\alpha a}{\hbar^2} \geq 1$.



$$z \simeq \frac{m\alpha a}{\hbar^2}\left(2z - 2z^2 + \dots\right) = \frac{2m\alpha a}{\hbar^2}\,z(1-z)$$

$$1-z = \frac{\hbar^2}{2m\alpha a} = \frac{a_c}{a} \qquad \left(a_c \equiv \frac{\hbar^2}{2m\alpha}\right)$$

$$z = 1-\frac{a_c}{a} = \frac{a-a_c}{a} > 0 \Rightarrow \varkappa = \frac{z}{a} = \frac{a-a_c}{a^2}$$